American International University- Bangladesh (AIUB)
Faculty of Engineering (EEE)
Subject: Engineering Ethics and Environmental Protection

Course Name:	Engineering Ethics and Environmental Protection	**Course Code:**	EEE 3107
Semester:Spring		**Submission date:**	**01-03-2026**
Item:	**CO1: Demonstrates individual responsibilities based on norms of engineering ethical practices. (P.h.2.A4)**		
Student Name:	**Student ID:**	**Student Name:**	**Student ID:**
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Department:	**CSE**	**Section:**	**I Group: 4**

Questions: Write down the **e-waste management internationally and Bangladesh perspective: Case 1-Mobile phones and its accessories (Group 2 and 3 will be different)** with appropriate references.

Group-1 Mobile Phones and its accessories
Group-2: Personal computers and its accessories
Group-3: Electrical components and accessories used in power generation stations and industries.

Criteria: Please include the following but not limited to (one example)……..

- **a)** Definition of E-waste
- **b)** Definition and types of Mobile phones or Personal computers or Electrical components and their accessories
- **c)** Materials used for Mobile phones or Personal computers or Electrical components and their accessories
- **d)** Lifetime of materials used for Mobile phones or Personal computers or Electrical components and their accessories
- **e)** Environmental impacts of materials used for Mobile phones or Personal computers or Electrical components and their accessories
- **f)** What are the existing e-waste's management ways of Mobile phones or Personal computers or Electrical components and their accessories in BD and globally.

Please add appropriate figures, tables where necessary with references from where you collected the information. Write in your own languages. Assignments will be checked with plagiarism software not more than 15% similarity. A minimum of 4 pages are necessary and take care of grammatical errors, typos.

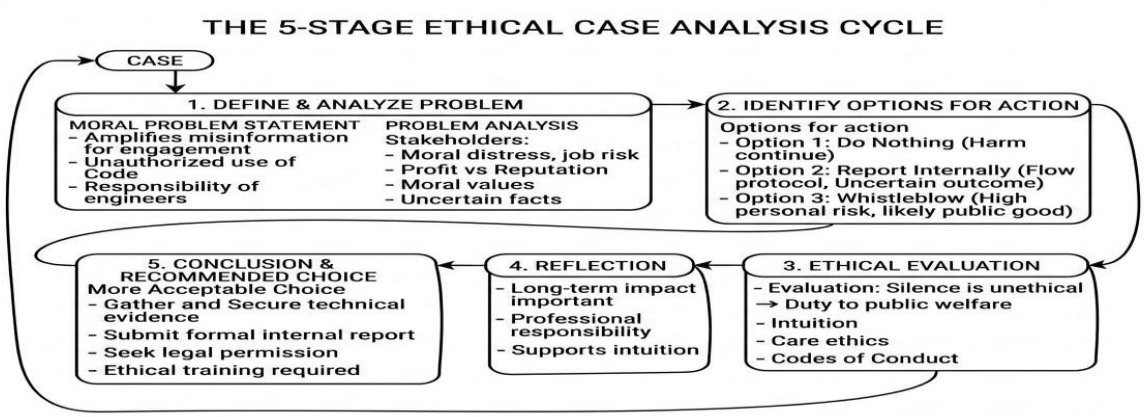
Rubric:

Category	Proficient [6]	Good [5]	Average [4-3]	Poor [2-1]	Secured Marks
Format of the code of conduct	Very well-structured of code of conduct with proper reference to the IEEE, ACM, NSPE, IEB etc.	Well-structured of code of conduct with proper reference to the IEEE, ACM, NSPE, IEB etc.	Acceptable format of code of conduct with some reference to the IEEE, ACM, NSPE, IEB etc.	Code of conduct without any reference to the IEEE, ACM, NSPE, IEB etc.	
Comprehension of the role of engineering in society (K7)	Comprehension of the role of engineering in society is stated clearly and described comprehensively, delivering relevant information	Comprehension of the role of engineering in society is stated, described, and clarified so that understanding is not	Comprehension of the role of engineering in society is stated, but description leaves some terms undefined, ambiguities	Comprehension of the role of engineering in society is stated without	

	necessary for full understanding.	seriously impeded by omissions.	unexplored, boundaries undetermined,	clarification or description.	
Depth of Knowledge (P1)	Specific position (perspective, hypothesis) is imaginative, considering the complexities of an issue. Limits of position (perspective, hypothesis) are acknowledged. Others' points of view and assumptions are synthesized within position (perspective, hypothesis).	Specific position (perspective, thesis/hypothesis) considers the complexities of an issue. Others' points of view and assumptions are acknowledged within position (perspective, hypothesis).	Specific position (perspective, hypothesis) acknowledges different sides of an issue.	Specific position (perspective, hypothesis) is stated, but is simplistic and obvious.	
Engineering responsibility to public safety and Environment (P2)	Engineering responsibility to public safety and environment is stated clearly and described comprehensively, delivering relevant information necessary for full understanding.	Engineering responsibility to public safety and environment is stated, described, and clarified so that understanding is not seriously impeded by omissions.	Engineering responsibility to public safety and environment is stated, but description leaves some terms undefined, ambiguities unexplored, boundaries undetermined,	Engineering responsibility to public safety and environment is stated without clarification or description.	
Importance of social, economic, and sustainable development (P6)	Importance of 'Safety' to social, economic and sustainable development is stated clearly and described comprehensively, delivering relevant information necessary for full understanding.	Importance of 'Safety' to social, economic and sustainable development is stated, described, and clarified so that understanding is not seriously impeded by omissions.	Importance of 'Safety' to social, economic and sustainable development is stated, but description leaves some terms undefined, ambiguities unexplored, boundaries undetermined,	Importance of 'Safety' to social, economic and sustainable development is stated without clarification or description.	
	For complete Similarity with other (Negative Marking will be imposed)				
Comments:			Total Marks (Out of 10):		

Solution:

Ethical cycle：



1. Introduction and Definition of E-waste

Electronic waste (e-waste) refers to discarded electrical and electronic devices that are no longer functional, obsolete, or unwanted. In the modern digital era, mobile phones and their accessories are one of the fastest-growing sources of e-waste due to rapid technological upgrades, short product lifecycles, and high consumer demand. Mobile phones contain both valuable materials and hazardous substances. Proper management of this waste is essential to protect public health, the environment, and ensure sustainable development. From an engineering ethics perspective, safe design, recycling, and disposal are moral responsibilities.

2. Types of Mobile Phones and Accessories

Mobile phones can be classified based on their functionality and complexity of hardware. Feature phones (also known as basic phones) are primarily used for voice calls and text messages; they have limited ability to connect to the internet. Smartphones, on the other hand, are very sophisticated with handheld computing devices with powerful processors, built-in cameras, built-in sensors, and internet connectivity. The most common accessories associated with mobile phones are: (1) lithium-ion batteries; (2) chargers and USB cables; (3) wired and wireless headphones; (4) portable backup batteries; (5) phone cases; (6) screen protectors.

3. Materials Used in Mobile Phones

Material Type	Examples	Typical Lifetime in Device
Precious Metals	Gold, Silver, Palladium, Copper	2-5 Years (Functional)
Hazardous Chemicals	Lead, Mercury, Cadmium, BFRs	Persistent for decades in soil
Battery Materials	Lithium, Cobalt, Nickel	2-3 Years
Polymers/Plastics	Polycarbonate, ABS	50-100+ Years

4. Environmental Impacts

Improper disposal of mobile phone e-waste can cause serious environmental pollution. Heavy metals from batteries contaminate soil and groundwater and get into the food chain, reducing the amount of food that can be grown. Burning of wires and plastics causes serious gases such as dioxins and furans to be released into the environment and can contribute to air pollution and global warming. Workers who recycle mobile phone e-waste in an informal manner are exposed to dangerous chemicals, which may lead to respiratory diseases, skin diseases, neurological injuries and reproductive health problems.

5. Global E-waste Management Practices

Countries like Switzerland, Germany and Japan utilize automated recycling facilities that return high value metals without exposing people to dangerous chemicals. Extended Producer Responsibility (EPR) is a set of rules for manufacturers to ensure their products are recovered and recycled when they have reached the end of their life. Many organizations provide consumers with ways to recycle phones by implementing take back programs. Many manufacturers are using recycled materials such as aluminium and cobalt as input into manufacturing new devices and adopting circular economy principles. The Right to Repair movement also allows phones to have longer lives by providing consumers with access to spare parts and repair manuals.

6. E-waste Management in Bangladesh

In Bangladesh alone, there is a large amount of mobile e-waste, & it is mostly processed illegally by the informal sector. 90–97% of e-waste is processed by small scrap dealers using unsafe methods such as burning and acid leaching to separate the metals from the waste. The government of Bangladesh introduced E-waste Management Rules in 2021, which include Extended Producer Responsibility and guidelines for safely recycling e-waste. While formal e-waste recycling companies are emerging, low collection rates still occur because many consumers store their old phones at home or sell them to informal collectors. The major obstacles preventing an increase in recycling rates are limited awareness, infrastructure, and monitoring.

9. Recommendations for Bangladesh

Develop formal e-waste recycling facilities equipped with state-of-the-art technologies. Conduct Formal Monitoring of the EPR (Extended Producer Responsibility) Policies. Increase Public Awareness Campaigns to educate the public about the safe disposal of mobile phones. Provide Training and Protective Equipment for Workers in the Recycling Industry. Encourage Telecom Companies to Develop Take Back Programs for Mobile Phones. Encourage Research into the Development of Eco-Friendly/Repairable Mobile Phone Designs.

Figure-1: E-waste recycling process

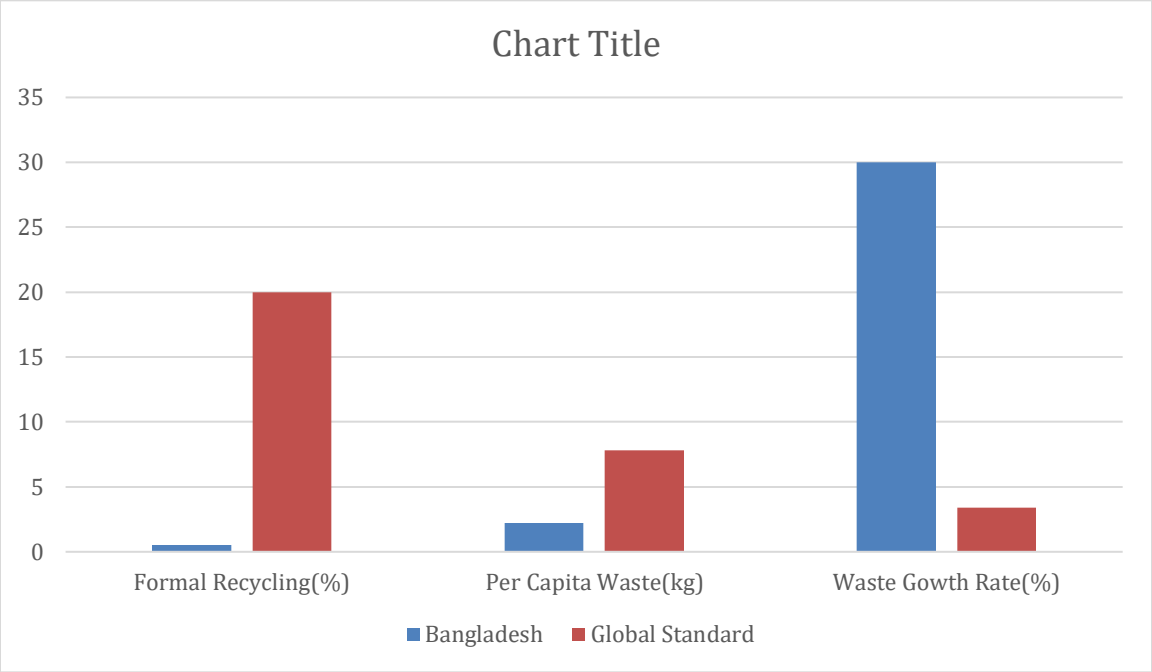


Figure-2: Comparison of Bangladesh and global e-waste management systems

11. Conclusion

Mobile phone e-waste management is both a technical and ethical challenge. While developed countries have advanced recycling systems and circular economy models, Bangladesh still relies heavily on informal practices. The introduction of national regulations is a positive step, but effective implementation, infrastructure development, and public awareness are required. Engineers have a moral responsibility to design sustainable products, ensure safe recycling, and protect public health and the environment.

12. References

[1] Government of the People's Republic of Bangladesh, Ministry of Environment, Forest and Climate Change, "Hazardous Waste (E-waste) Management Rules, 2021," Dhaka, Bangladesh, 2021.

[2] United Nations University (UNU), International Telecommunication Union (ITU), and International Solid Waste Association (ISWA), The Global E-waste Monitor 2020: Quantities, Flows and the Circular Economy Potential, Bonn/Geneva/Rotterdam, 2020.

[3] European Parliament and Council of the European Union, "Directive 2012/19/EU on Waste Electrical and Electronic Equipment (WEEE)," Official Journal of the European Union, 2012.

[4] United Nations Environment Programme (UNEP), Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal, 1989.

[5] S. Widmer, R. Oswald-Krapf, D. Sinha-Khetriwal, M. Schnellmann, and H. Böni, "Global perspectives on e-waste," Environmental Impact Assessment Review, vol. 25, no. 5, pp. 436–458, 2005.

[6] United Nations Environment Programme (UNEP), E-waste Management Manual, Osaka/Shiga, Japan, 2007.